RideWire Job Aid — C1032 / C1034

Wheel Speed Sensor Circuit Open or Shorted (Front or Rear).

*This job aid is AI-assisted research support compiled from real Harley-Davidson service bulletins, diagnostic manuals, and owner/technician forum reports, cross-checked by three independent AI models. It is **not** a certified repair procedure and does not replace the vehicle's factory service manual. Always verify against manufacturer documentation before performing repairs.*

Platform

Harley-Davidson ABS-equipped Touring and related models (front = C1032, rear = C1034)

Code Description[1]

Wheel Speed Sensor Circuit Open or Shorted (Front or Rear). Per a Harley-Davidson electrical diagnostic manual excerpt reproduced by an owner, the DTC sets when the ABS module detects: a short to ground on either the high or low side circuit of the wheel speed sensor; an open on either the high or low side circuit; a short to voltage on the low side causing voltage above 0.79V; or the wheel speed sensor itself being open. When set, the ABS module disables ABS function and illuminates the ABS indicator.

Likely Causes (verified sources)

- A broken or pinched wheel speed sensor wire, commonly at the steering neck/fork area where the harness can stretch or rub when the handlebars are turned to full lock
- Open circuit in either of the wheel speed sensor's high- or low-side wires, or a short to ground in the sensor wiring
- A failed/open wheel speed sensor itself, or a short to battery voltage on the low-side circuit

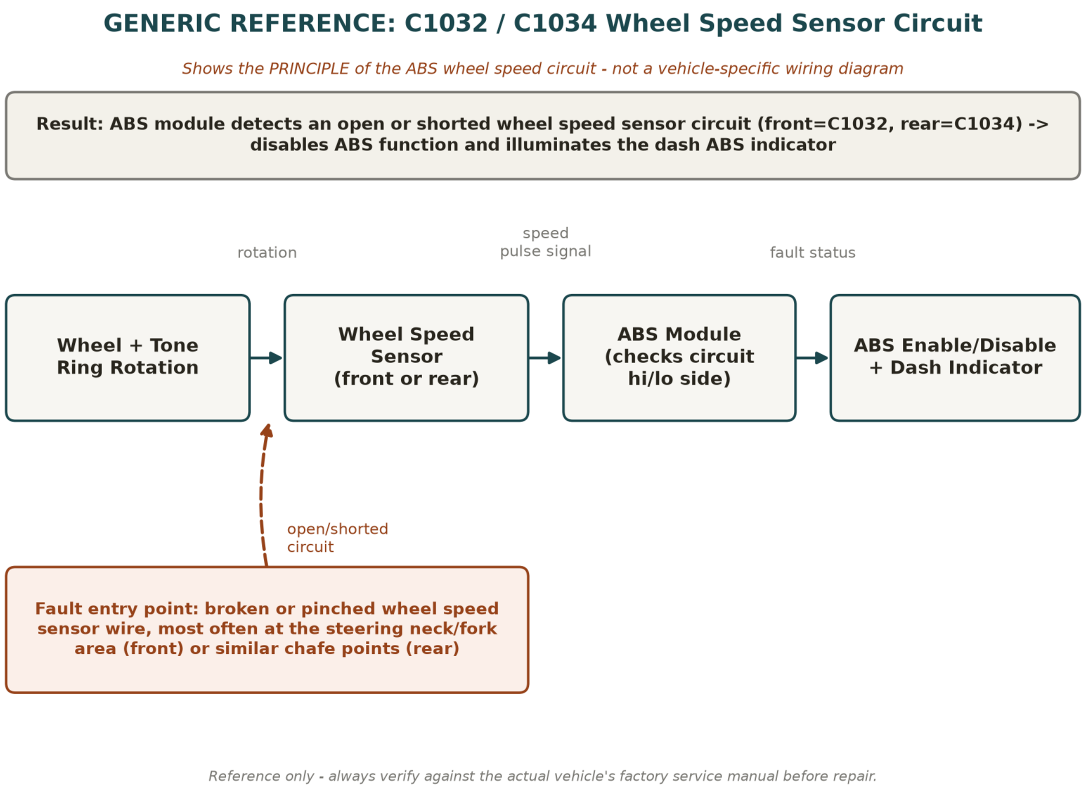
Recommended First Check[2]

Per the Harley-Davidson diagnostic manual procedure, first inspect the wheel speed sensor wiring harness for physical damage (especially at the steering-neck/fork area for the front sensor), then use a breakout box / ohmmeter to check continuity between the ABS module connector and the wheel speed sensor connector; a broken or intermittent wire (frequently found at the neck where the harness contacts the frame at steering lock) is the most commonly reported real-world cause.

Source confidence: Moderate-strong: the primary source is a hdforums.com thread that reproduces what its poster states is the verbatim Harley-Davidson diagnostic-manual page for 'DTC C1032, C1034: WHEEL SPEED SENSOR CIRCUIT OPEN OR SHORTED (FRONT OR REAR)', including specific tool part numbers (HD-41404-B, HD-48642) that support it being genuine dealer service literature rather than a paraphrase. This is independently corroborated by a second, separate hdforums.com thread describing a real-world C1032 case with a broken sensor wire at the steering neck as the root cause, and further supported contextually by an official NHTSA-hosted Harley-Davidson safety recall notice (Campaign 0175, RCMN-19V843-1027) confirming that faulty wheel-speed-sensor signals are a documented, safety-relevant fault class on Harley-Davidson ABS/traction-control-equipped motorcycles (Trike platform). IMPORTANT CAVEAT: an NHTSA-hosted bulletin literally titled 'M16-12-02 ABS Wheel Speed Sensor Diagnostics' with matching C1032/C1033/C1034/C1035 code numbers was initially found and looked like a strong third source, but on closer inspection it applies to the MV-1 wheelchair-accessible vehicle (Mobility Ventures LLC), NOT a Harley-Davidson motorcycle -- it was deliberately excluded from the sources list to avoid conflating an unrelated vehicle's identical-looking code numbers with Harley's own C-code family. Because full manufacturer PDF diagnostic-manual pages for C1032 could not be independently located on a second fully separate hosting source, this code is rated moderate-strong rather than fully verified against two wholly independent primary documents.

Reference Schematic

Generic principle-level diagram — not a manufacturer-exact wiring diagram for any specific make/model/year. Always verify against the factory service manual.



Multi-AI Consensus (3 independent models)

“Agreement” below is a plain keyword-overlap check on each model's own wording for this single query — it is NOT a validated machine-learning accuracy score and has not been checked against confirmed repair outcomes. Read each model's own words in the table for the actual reasoning.

Model	Primary Root Cause	Urgency	Recommended First Check
Claude Sonnet 5	Damaged, corroded, or disconnected wheel speed sensor wiring/connector (open or shorted circuit)	high	Visually inspect and ohm-test the affected wheel speed sensor and its wiring harness/connector (front for C1032, rear for C1034) for continuity, correct resistance, corrosion, and physical damage, and check sensor air gap to the tone ring
GPT-5.6 Terra	Damaged or disconnected wheel-speed-sensor wiring or connector	high	Disconnect the affected front (C1032) or rear (C1034) wheel-speed-sensor connector and inspect the lead and terminals, then check the sensor circuit for continuity and shorts while flexing the harness.
Gemini 3.1 Pro	Damaged wheel speed sensor wiring or faulty sensor	high	Visually inspect the wheel speed sensor wiring for chafing or breaks, and check the connector for corrosion.

Root cause agreement: 3 of 3 models shared the keyword(s) ['damaged'] in their primary root cause

Urgency values reported: high, high, high (all 3 models agree)

Sources

1. https://www.hdforums.com/forum/touring-models/547091-anti-lock-brake-light-is-staying-on.html
2. https://www.hdforums.com/forum/touring-models/1281928-abs-fault-1032-a.html
3. https://static.nhtsa.gov/odi/rcl/2019/RCMN-19V843-1027.pdf

Superscript 1-2 in the text above refer to the same verified source set backing the code description and first-check procedure, drawn from the sources listed here.